Module 2 Quiz: Conditional Probability, Bayes' Rule, Likelihood, Distributions, and Asymptotics

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Due Aug 2, 11:59 PM IST **Attempts** [illegible]

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1. You meet a person at the bus stop and strike up a conversation. In the conversation, the person gives the strange answer that at least one of his two children is a girl. Thinking on this, you decide to do a probability calculation. Assuming only that genders are iid with 50% probability each, what is the chance of a two child family having two girls given the information that at least one is a girl? **1 / 1 point**

Correct

2. A web site (www.medicine.ox.ac.uk/bandolier/band64/b64-7.html) for home pregnancy tests cites the following: "When the subjects using the test were women who collected and tested their own samples, the overall sensitivity was 75%. Specificity was also low, in the range 52% to 75%." Suppose a subject has a positive test. Assume the lower bound for the specificity. What number is closest to the multiplier of the pre-test odds of pregnancy to obtain the post-test odds of pregnancy given a positive test result? **1 / 1 point**

Correct

3. A web site (www.medicine.ox.ac.uk/bandolier/band64/b64-7.html) for home pregnancy tests cites the following: "When the subjects using the test were women who collected and tested their own samples, the overall sensitivity was 75%. Specificity was also low, in the range 52% to 75%." Assume the lower value for the specificity. Suppose a subject has a positive test and that 30% of women taking pregnancy tests are actually pregnant. What number is closest to the probability of pregnancy given the positive test? **1 / 1 point**

Correct

4. Let $X_1 \ldots X_K$ be independent Poisson counts with means $t_i \lambda$ for some known value t_i. Recall the Poison mass function with mean μ is **1 / 1 point**

$$\frac{\mu^x e^{-\mu}}{x!}$$

for $x = 0, 1, \ldots$. What is the maximum likelihood estimate for λ?

Correct

5. Suppose that a person is flipping a biased coin with success probability p. She flips the coin 10 times yielding 1 head. Consider two possibilities: 1) the person planned on flipping the coin ten times and got one head, 2) the person planned to flip the coin until the first head and it took ten times. What can you say about the likelihood in these two circumstances? **1 / 1 point**

Correct

6. Let X be a uniform random variable with support of length 1, but where we don't know where it starts. So that the density is **1 / 1 point**

$$f(x) = 1$$

for $x \in (\theta, \theta + 1)$ and 0 otherwise. We observe a random variable from this distribution, say x_1. What does the likelihood look like?

Correct

7. Suppose that diastolic blood pressures (DBPs) for men aged 35-44 are normally distributed with a mean of 80 (mm Hg) and a standard deviation of 10. What is the probability that a random 35-44 year old has a DBP less than 70? **1 / 1 point**

Correct

8. Brain volume for adult women is normally distributed with a mean of about 1,100 cc for women with a standard deviation of 75 cc. About what brain volume represents the 95th percentile? **1 / 1 point**

Correct

9. Return to the previous question. Brain volume for adult women is about 1,100 cc for women with a standard deviation of 75 cc. Consider the sample mean of 100 random adult women from this population. Around what is the 95th percentile of the distribution of that sample mean? **1 / 1 point**

Correct

10. Recall from an earlier question. Suppose that diastolic blood pressures (DBPs) for men aged 35-44 are normally distributed with a mean of 80 (mm Hg) and a standard deviation of 10. What's the probability that in a random sample of 5 subjects, 4 or more have DBPs more than 90? **1 / 1 point**

Correct

11. Consider two sample means, $\bar{X}_1$ and $\bar{X}_2$ from a the same normal population having mean zero and standard deviation σ. The first sample mean is based on n_1 observations while the second is based on n_2 observations. Take any percentile, say $100 \times \alpha$, for the distribution of sample means based on the two sample sizes excluding the median. What is the ratio of the two percentiles where it is the percentile of the first group divided by the second? **1 / 1 point**

Correct

12. You flip a fair coin 5 times, what's the probability of getting 4 or 5 heads? **1 / 1 point**

Correct

13. The respiratory disturbance index (RDI), a measure of sleep disturbance, for a specific population has a mean of 15 (sleep events per hour) and a standard deviation of 10. They are not normally distributed. Give your best estimate of the probability that a sample mean RDI of 100 people is between 14 and 16 events per hour? **1 / 1 point**

Correct

14. Consider a standard uniform density. The mean for this density is .5 and the variance is 1 / 12. You sample 1,000 observations from this distribution and take the sample mean, what value would you expect it to be near? **1 / 1 point**

Correct

15. Consider a standard uniform density. The mean for this density is .5 and the variance is 1 / 12. You sample 1,000 sample means where each sample mean is comprised of 100 observations. You take the standard deviation of the 1,000 sample means. About what number would you expect it to be? **1 / 1 point**

Correct

16. Consider a standard uniform density. The mean for this density is .5 and the variance is 1 / 12. You sample 1,000 sample variances where each sample variance is comprised of 100 observations. You take the average of the 1,000 sample variances. What number would you expect that to be near? **1 / 1 point**

Correct